

NAVLEEN SAINI

FULL STACK DEVELOPER

431-277-9876 | navanmol777@gmail.com | [GitHub](#) | [Portfolio](#)

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PORFESSIONAL SUMMARY

Applied Computer Science student with strong foundation in algorithms, data structures, and backend development. Experienced in building server-side applications using Node.js and Express, implementing custom data structures in C++, and designing relational databases using SQL. Interested in software engineering and intelligent systems.

TECHNICAL SKILLS

Programming Languages: C++, Python, JavaScript

Backend: Node.js, Express.js

Core CS: Data Structures, Algorithms, OOP, Time & Space Complexity Analysis

AI Concepts: Search Algorithms (BFS, DFS, A*, heuristic search)

Database: SQL, Relational Database Design

Tools & Design: Git, Figma

Other: Physical Computing, Hardware-Software Integration

PROJECTS

Backend-Focused Web Application (Node.js, Express, JavaScript) | [GitHub \(backend\)](#) | [GitHub \(frontend\)](#)

- Implemented user authentication and authorization using secure password hashing (bcrypt, salting) and protected routes.
- Developed custom middleware for request validation and access control.
- Integrated MongoDB using Mongoose for schema modeling and database interaction.
- Applied input validation and server-side error handling to ensure data integrity and security.
- Structured project using layered architecture (routes, controllers, models) for maintainability.

Autonomous Safety Robot (Arduino, C++) | [Portfolio \(robot video\)](#)

- Programmed robot with real-time control and obstacle avoidance.
- Applied optimized C++ functions and signal smoothing for faster, smoother operation.

Fast search AI research (on going | python)

- Implementing heuristic search algorithms including A* and binary search in Python.
- Analyzing algorithm efficiency and exploring integration of AI search methods into backend systems.

Simple Portfolio deployment (html, CSS) | [Portfolio](#)

Written code basic html and CSS for portfolio and deployed using GitHub.

EDUCATION AND CERTIFICATION

B.sc in Applied Computer Science | University of Winnipeg (2023 - 2026)

Major GPA: 3.7 | CGPA: 3.4

EXPERIENCE

Made Portfolios for classmates | Html and CSS (2024 - 2026)

Overnight manger | Tim Hortons (2023 - 2026)

Teaching Assistant | NYC ACS (2022 - 2023)